NIHR | National Institute for Health and Care Research

Summary

Reference Number	35551
Lead Applicant	Dr Kathryn Biddle
Research Title	Doctoral Fellowship R12
Plain English Summary	

Background
Sarcoidosis is a disease that causes lumps called "granulomas" to form in the body. These granulomas cause inflammation and organ damage. Sarcoidosis is rare, but it is important to understand because it can cause death.

The problem
Sarcoidosis affects patients in very different ways. In some patients, sarcoidosis is very mild and gets better on its own. In other patients, sarcoidosis can be deadly. It is very difficult to predict which patients will develop severe disease. It's also hard to predict which patients will die from their disease. Therefore, doctors struggle to know which patients need treatment. There haven't been many clinical trials in sarcoidosis and we don't know which treatment is best.

Sarcoidosis can affect any part of the body. PET-CT is a special type of medical scan which can find affected organs and guide treatment. Doctors at different hospitals use PET-CT scans in different ways. If we knew how best to use PET-CT scans, they could tell us which patients are at the greatest risk of severe disease.

Aim of my research
The aim of my research is to identify which patients are at risk of severe sarcoidosis. We want to know how factors such as age, sex, and ethnicity affect the risk of severe disease. We also want to know how the pattern of disease on PET-CT scans affects the patients' risk.

My research will help:

- Find patients who need early treatment. This will reduce the risk of organ damage and death.
- Identify patients who qualify for clinical trials of new treatments. This will drive further research into sarcoidosis, with the hope of finding new effective medications.
- Target resources to groups of people who need them the most. This will help NHS managers to allocate healthcare services effectively.

Design and methods
I will divide my research into three parts.
Phase 1: We will collect evidence from medical records in GP practices. We will do this using a secure system called OpenSAFELY. We will look at patients with sarcoidosis and compare those with mild and severe disease. We will look at their age, sex, ethnicity. We will also consider their other health conditions and pattern of sarcoidosis. We will determine whether there are any factors that increase the risk of death or organ damage.
Phase 2: We will train an artificial intelligence (AI) model. It will measure disease in the heart and lungs on PET-CT scans. We will test the model to make sure that it is accurate.
Phase 3: We will look at patients with sarcoidosis over a one-year follow-up period. All patients will have a PET-CT at baseline. They will do this as part of their routine clinical care. We will use the AI model to find PET-CT scan features that predict severe disease one year later.

Patient and public involvement
I have worked with the charity Sarcoidosis UK to speak to different patients. I conducted interviews and a focus group to understand the impact of sarcoidosis. Six patient representatives will continue to guide the project. They will help to design the study and analyse the results. Patient input will ensure that research focuses on benefiting patients.

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	Dissemination My patient panel will help ensure that my results reach as many people as possible. Many approaches will be used: • Plain English summaries of all publications. • Newsletters of outputs, which can be disseminated through charities including Sarcoidosis UK. • Social media outputs, including Facebook posts. • Public engagement events, organised in conjunction with Sarcoidosis UK.

Pre-submission

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

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1. Application Summary Information

Host organisation (which will administer any award)
King's College London

Partner organisation (if applicable)	King's College Hospital NHS Foundation Trust

Application Type

Strategic Theme	N/A

Partnership Fellowship	N/A

WTE % Please select which WTE option you would prefer
1.00

Duration Doctoral Fellowship is a 3 year award and based on your WTE preference the duration in months is calculated as follows
36

Research Title	Doctoral Fellowship R12

Research Type	Cohort study, Epidemiological, Retrospective review

Proposed start date if grant awarded	01/06/2025

Pre-submission

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

The 'Publication Record' section of the form is automatically populated from publications added into the "My Research Outputs" page of your ARAMIS account.

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class	Department
09/2020	06/2021	PGCert	Healthcare Research Skills and Methods	United Kingdom (England)	St Georges University	Pass	Medical Sciences
09/2018	06/2019	MSc	Integrated Immunology	United Kingdom (England)	The University of Oxford	Pass	Nuffield Department of Surgical Sciences (NDS)
09/2010	06/2016	MBBChir	Medicine	United Kingdom (England)	The University of Cambridge	Distinction	School of Clinical Medicine
09/2010	06/2013	MA	Neuroscience		The University of Cambridge	First	Clare College

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
10/2023	-	Rheumatology Specialist Trainee Year 5	Guys and St Thomas' NHS Foundation Trust	0
10/2022	10/2023	Rheumatology Specialist Trainee Year 4	Kingston Hospital NHS Foundation Trust	0
08/2019	08/2022	NIHR Academic Clinical Fellow in Rheumatology and Internal Medicine	St Georges University NHS Foundation Trust	25
08/2018	08/2019	Trust Speciality Doctor (General Medicine)	Countess of Chester Hospital	0
08/2016	08/2018	Foundation Trainee	Countess of Chester Hospital	0

Research Grants held

Publication Record
(include the list of authors, article title, journal name, and volume)
This information can be edited via the "My Research Outputs" page of your ARAMIS account.

Publications
Biddle, K., Agaoglu, E., Brent, G., Adam, J., Nockels, R., Loveridge, A., Bazari, F., Natkunarajah, J. and Al-Shakarchi, I.. (2024) 'Two challenging cases of anti-MDA-5 dermatomyositis with rapidly progressive interstitial lung disease.', *Oxford Medical Case Reports*, 2024(6).
Ahmed, L., Biddle, K., Blundell, A., Koushesh, S., Kiely, P., Mein, G., Sedgwick, P. and Sofat, N.. (2023) 'Assessing the effects of distinct biologic therapies on rheumatoid arthritis pain by nociceptive, neuropathic and nociplastic pain components: a randomised feasibility study',

Pre-submission

Publication Record

(include the list of authors, article title, journal name, and volume)
This information can be edited via the "My Research Outputs" page of your ARAMIS account.

Publications
Biddle, K., Blundell, A. and Sofat, N.. (2023) 'Understanding Clinical Research: an introduction', *Understanding Clinical Research: an introduction* , (ISBN 1914961269).
Biddle, K., Nathan, J. and Sandhu, V.. (2023) 'Treatment of gangrenous digit-threatening paraneoplastic acrocyanosis with vasodilator therapy.', *BMJ Case Reports*, 16(2).
Biddle, K. and Kaul, A.. (2022) 'Lingual Raynaud's phenomenon: a rare presentation.', *BMJ Case Reports*, 15(11).
Biddle, K., Taylor, MD., Linstead, SE. and Kiely, PDW.. (2022) 'The Utility of Myositis Specific Antibodies in Clinical Practice.', *The journal of applied laboratory medicine*, 7(5).
Biddle, K., White, J. and Sofat, N.. (2022) 'What is the full potential of baricitinib in treating patients with COVID-19?', *Expert Review of Clinical Immunology*, 18(6).
Rotondo, F., Biddle, K., Chen, J., Ferencik, J., d'Esneval, M. and Milton, AL.. (2021) 'Lack of Effect of Propranolol on the Reconsolidation of Conditioned Fear Memory due to a Failure to Engage Memory Destabilisation.', *Neuroscience*, 480.
Biddle, K. and Sofat, N.. (2020) 'Understanding the Mechanisms of Pain in Rheumatoid Arthritis', *Rheumatoid Arthritis - Other Perspectives towards a Better Practice*, (ISBN 978-1-83962-533-6).
Ahmed, SH., Biddle, K., Augustine, T. and Azmi, S.. (2020) 'Post-Transplantation Diabetes Mellitus.', *Diabetes therapy : research, treatment and education of diabetes and related disorders*, 11(4).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/06/2017	Countess of Chester Hospital	Runner Up FY1 Doctor of the Year - Nominated for being a "great support to colleagues" and for "going above and beyond the expectations of an F1"	Recipient
01/06/2015	University of Cambridge	Gresham Prize in Pathology (Awarded for Top Results in Pathology Examinations)	Recipient
01/06/2015	University of Cambridge and Harvard	Clare-Massachusetts General Hospital Studentship - I was awarded a studentship with the opportunity to spend seven weeks in Boston for my elective medical placement.	Recipient
01/06/2014	University of Cambridge	William Harvey Studentship Award (Awarded for Results in Fourth Year Examinations)	Recipient
01/06/2013	Clare College, University of Cambridge	College Prize and Foundation Scholarship (First Class Examination Results)	Recipient
01/06/2012	Clare College, University of Cambridge	College Foundation Scholarship (First Class Examination Results)	Recipient
01/06/2012	Clare College, University	William Butler Prize for	Recipient

Pre-submission

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
	of Cambridge	Medical Sciences (Best in College)	
01/06/2011	Clare College, University of Cambridge	College Prize in Examinations (First Class Examination Results)	Recipient

ORCID iD 0000-0001-5267-200X

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3. Applicant Research Background

Applicant Professional Background	Medically Qualified

Specialty (if Medically Qualified)
Rheumatology

Do you have a GMC number?	Yes

GMC number Please supply a valid GMC number	7518176

Please describe your research career to date

Career to date

My career experiences have reinforced my desire to pursue a career as a leading clinical academic in Rheumatology. I completed my undergraduate medical studies and an intercalated degree at the University of Cambridge, earning yearly academic prizes. My Foundation Training took place at the Countess of Chester Hospital, followed by an MSc in Immunology at the University of Oxford. Subsequently, I secured an NIHR Academic Clinical Fellowship (ACF) with run-through specialty training in Rheumatology. Currently, I am an ST5-level trainee in the South London programme.

Research Experience

I developed a passion for research during my intercalated degree with a project investigating the effect of propranolol on the reconsolidation of conditioned fear memories. This involved animal model work, laboratory work and a literature review, resulting in a second-author publication in *Neuroscience*.

I was first drawn to a career in rheumatology during my Foundation Training. During this time, I audited the local management of inflammatory arthritis and led a pilot trial on the efficacy of dietary intervention for gastroparesis in patients with systemic sclerosis. I presented these projects at the British Society for Rheumatology (BSR) annual conference. I also furthered my skills in critical appraisal by co-authoring a review article on post-transplantation diabetes mellitus.

My MSc in Immunology enhanced my understanding of basic science and critical appraisal. During this time, I undertook a research project investigating the effect of talarozole on inflammatory signalling in chondrocytes. This involved laboratory work (chondrocyte isolation, culture and stimulation and Western blotting) and data analysis.

My primary NIHR ACF research project was a feasibility trial investigating the effect of different biological medications on pain outcomes in participants with rheumatoid arthritis. I consented participants for recruitment, conducted study visits and provided supervision to the research nurses and research fellows; enhancing my

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leadership and management skills. I also performed data analysis, using SPSS and GraphPad. Although participant recruitment was lower than expected due to the COVID-19 pandemic, the work was published and I was proud to be a co-author.

The COVID-19 pandemic redirected my ACF research to studying the clinical outcomes of patients with autoimmune conditions who developed COVID-19. During this work, I submitted applications for ethical approval, conducted trial visits, performed laboratory work (including PBMC isolation, ELISAs and ELISpots) and analysed trial data. I presented my findings at national and international conferences.

During my ACF, I authored three literature reviews, two case reports and a book chapter, improving my skills in scientific writing. I was the first author of a book introducing clinical research concepts to medical professionals. I wrote chapters on statistics, clinical study design and ethical and governance considerations in research. I also delivered an oral presentation at the BSR Case-Based Conference and completed the PGCert in Healthcare Research Skills and Methods at St George's University with modules in statistics, research methods and critical appraisal.

I am a current member of the BSR working group for the ANCA-associated vasculitis guidelines. I have attended working group meetings, completed a systematic literature review, written consensus statements and drafted the final document. Throughout this process, I have worked closely with academic experts at leading UK centres, highlighting the importance of networking and collaboration in research.

Research Experience in Sarcoidosis

My rheumatology training sparked an interest in multi-system conditions, particularly sarcoidosis. Whilst caring for patients with sarcoidosis, I became aware of the paucity of research into this condition, when compared to other inflammatory diseases. Despite being associated with significant morbidity and mortality, I noticed a significant lack of epidemiological and trial data in this area.

Working with Dr Bechman and Professor Galloway, I performed a systematic review of biological medications in sarcoidosis. Only nine RCTs were identified, highlighting an unmet research need. This work was presented at the European Alliance of Associations for Rheumatology (EULAR) Congress and I have written a manuscript that is currently under submission. More recently, we have used population-wide primary care data to describe the epidemiology of sarcoidosis in England, highlighting a four-fold increase in prevalence over the last two decades. We have submitted this work to the American College of Rheumatology conference, and are currently drafting a manuscript for submission for publication.

To foster collaboration with charity representatives, I am working with Sarcoidosis UK to write patient information leaflets. I am also organising a national multi-disciplinary sarcoidosis study day at King's College London.

Management Experience

As an ST4 trainee, I served as Royal College of Physicians (RCP) Associate Clinical Tutor, liaising between medical trainees and the RCP College Tutor and organising regular postgraduate teaching, including a mock exam for medical trainees. As the current Chief Rheumatology Registrar at Guy's and St Thomas' Hospital, I attend departmental meetings, organise medical student teaching, and oversee the registrar rota.

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Teaching Experience

During medical school, I supervised junior students in clinical examination and pathology. In Foundation Training, I designed and organised simulated on-call sessions for final-year medical students and presented the feedback as posters at National and International Conferences.

Throughout training, I have developed my expertise in medical education; authoring rheumatology-themed questions for nationwide educational platforms and delivering a pan-London session on “Rheumatology Spot Diagnoses” that was attended by 315 doctors. I continue to teach regularly.

Impact of Fellowship

Sarcoidosis is an under-researched condition associated with significant mortality and morbidity. If awarded an NIHR doctoral research fellowship, I aim to develop methods to identify patients with sarcoidosis at high risk of severe disease. My research will provide objective clinical and radiological measurements of disease activity, guiding clinical trial design on a global scale. Improved identification of patients with severe disease will guide referral to specialist centres and prescription of second-line immunosuppressive therapies, improving patient quality of life and reducing healthcare burdens.

This fellowship would enable me to pursue my career goal of becoming a research leader in the field of sarcoidosis. The skills that I gain during this research training programme will put me in a strong position to submit a successful application for an NIHR Clinical Lectureship and to continue my work in sarcoidosis.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
No

Contextual Factors
Please use this section to detail any factors that may have impacted the development of your research career or outputs from your research career to date. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research activities (including research outputs), due to disability or to caring responsibilities. This may include physical or mental health related impairments or conditions, which have impacted your opportunities to undertake research activities
- Reduced opportunities for career support e.g., mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career

Please also use this section to detail any other factors that may have impacted your research career not listed in

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the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, contextual factors should be significant, and relevant. NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with the General Data and Protection Regulations (GDPR).
My NIHR ACF project was interrupted due to the COVID-19 pandemic. During my ACF, I lost research time as I was asked to continue medical on-call duties.

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4. Plain English Summary of Research

Background

Sarcoidosis is a disease that causes lumps called “granulomas” to form in the body. These granulomas cause inflammation and organ damage. Sarcoidosis is rare, but it is important to understand because it can cause death.

The problem

Sarcoidosis affects patients in very different ways. In some patients, sarcoidosis is very mild and gets better on its own. In other patients, sarcoidosis can be deadly. It is very difficult to predict which patients will develop severe disease. It’s also hard to predict which patients will die from their disease. Therefore, doctors struggle to know which patients need treatment. There haven’t been many clinical trials in sarcoidosis and we don’t know which treatment is best.

Sarcoidosis can affect any part of the body. PET-CT is a special type of medical scan which can find affected organs and guide treatment. Doctors at different hospitals use PET-CT scans in different ways. If we knew how best to use PET-CT scans, they could tell us which patients are at the greatest risk of severe disease.

Aim of my research

The aim of my research is to identify which patients are at risk of severe sarcoidosis. We want to know how factors such as age, sex, and ethnicity affect the risk of severe disease. We also want to know how the pattern of disease on PET-CT scans affects the patients’ risk.

My research will help:

- Find patients who need early treatment. This will reduce the risk of organ damage and death.
- Identify patients who qualify for clinical trials of new treatments. This will drive further research into sarcoidosis, with the hope of finding new effective medications.
- Target resources to groups of people who need them the most. This will help NHS managers to allocate healthcare services effectively.

Design and methods

I will divide my research into three parts.

Phase 1: We will collect evidence from medical records in GP practices. We will do this using a secure system called OpenSAFELY. We will look at patients with sarcoidosis and compare those with mild and severe disease. We will look at their age, sex, ethnicity. We will also consider their other health conditions and pattern of sarcoidosis. We will determine whether there are any factors that increase the risk of death or organ damage.

Phase 2: We will train an artificial intelligence (AI) model. It will measure disease in the heart and lungs on PET-CT scans. We will test the model to make sure that it is accurate.

Phase 3: We will look at patients with sarcoidosis over a one-year follow-up period. All patients will have a PET-CT at baseline. They will do this as part of their routine clinical care. We will use the AI model to find PET-CT scan features that predict severe disease one year later.

Patient and public involvement

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I have worked with the charity Sarcoidosis UK to speak to different patients. I conducted interviews and a focus group to understand the impact of sarcoidosis. Six patient representatives will continue to guide the project. They will help to design the study and analyse the results. Patient input will ensure that research focuses on benefiting patients.

Dissemination

My patient panel will help ensure that my results reach as many people as possible.

Many approaches will be used:

- Plain English summaries of all publications.
- Newsletters of outputs, which can be disseminated through charities including Sarcoidosis UK.
- Social media outputs, including Facebook posts.
- Public engagement events, organised in conjunction with Sarcoidosis UK.

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5. Scientific Abstract

Research question

Which patients with sarcoidosis are most at risk of severe disease?

Background

Sarcoidosis is a multisystem and clinically heterogeneous condition. It is associated with high mortality rates, primarily due to respiratory or cardiac involvement. The prevalence of sarcoidosis is rising, with our preliminary data showing a four-fold increase over the last 20 years in the UK.

Corticosteroids are first-line therapy in sarcoidosis but are associated with significant side effects and relapse when tapered. There is limited evidence to guide which patients will benefit from second or third-line immunosuppressive medications. Treatment is therefore non-standardised and often based on expert opinion.

In sarcoidosis, affected body organs show a particular affinity for the 18F-FDG tracer used in PET-CT, meaning that this imaging modality has a unique ability to quantify organ involvement. Despite this, objective measures of sarcoidosis activity on PET-CT do not exist.

There is a need to develop robust methods to identify patients at risk of severe forms of disease. This will enable appropriate referral to specialist centres, early consideration of second-line treatments and standardisation of recruitment into clinical trials.

Aims and objectives

My research plan has 3 phases:

1. To identify demographic and clinical risk factors for adverse outcomes in patients with sarcoidosis.
2. To develop a deep-learning model capable of quantifying sarcoidosis activity in the heart and lungs on PET-CT.
3. To identify PET-CT correlates of disease activity and future disease outcomes.

Methods (timelines of delivery)

Phase 1 (0-12 months): Using the nationwide OpenSAFELY platform, I will analyse clinical and demographic risk factors for adverse outcomes, including end-organ damage, hospitalisation and death. Regression models will be used to correlate predictors of interest with adverse outcomes.

Phase 2 (6-18 months): Using PET-CT performed in patients with confirmed sarcoidosis, I will develop a novel deep-learning algorithm to quantify and characterise disease burden and pattern on PET-CT, focusing on uptake in the heart and lungs.

Phase 3 (12-36 months): I will perform an observational trial in patients with biopsy-confirmed sarcoidosis with PET-CT performed at baseline. Using the deep-learning algorithm developed in Phase 2, I will use regression models to correlate disease burden with contemporary disease activity and clinical outcome at one year.

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Patient representatives will be involved at all stages of study design and analysis, ensuring that my research remains patient-focused.

Anticipated impact and dissemination

My research will allow non-expert clinicians to identify patients at risk of life-threatening sarcoidosis, who should be considered for referral to specialist centres and treatment with second-line immunosuppressive medications. This will improve the care of patients with sarcoidosis, reducing hospitalisations and death. My research will also inform the design of clinical trials of immunosuppressive medications for sarcoidosis on a global scale. I will disseminate my findings through publications, presentations and collaboration with colleagues from relevant medical specialities e.g. cardiology and respiratory medicine. The expertise of my PPI panel will guide dissemination to patients, carers and the wider public.

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6. Detailed Research Plan

What is the problem being addressed?

Sarcoidosis is an inflammatory condition characterised by granuloma formation in organs, most commonly the lungs and lymphatic system. Its clinical presentation can vary markedly, from mild and self-limiting, to severe organ damage and death[1]. Sarcoidosis is associated with an increased risk of mortality, largely due to respiratory or cardiac involvement[2,3]. It is a rare disease and therefore research into sarcoidosis is often neglected.

Throughout this application, I have worked closely with patients and gained an appreciation of the enormous impact of a diagnosis of sarcoidosis on the individual. Not only is the condition associated with burdensome symptoms, treatment options are limited and often lead to significant side effects. Our preliminary data shows that about 22,000 people are living with sarcoidosis in England. These patients represent an underserved population in urgent need of evidence-based improvements in their care.

Research into sarcoidosis is often underfunded due to a lack of interest from pharmaceutical companies due to disease rarity. Corticosteroids remain the first-line therapeutic option but are associated with significant side effects including hypertension, diabetes, weight gain, osteoporosis, psychosis and infection[4]. Some patients with sarcoidosis do not respond to corticosteroids and require second-line therapy e.g. methotrexate[5]. Second-line immunosuppressive agents are effective treatments that can reduce steroid burden[6]. Although experts advocate for earlier use of steroid-sparing agents, they are prescribed for less than half of treatment-requiring patients[7].

The focus of my research proposal is to understand which patients are at the highest risk of developing severe disease. This will allow the identification of a high-risk cohort of patients, who should be considered for second-line immunosuppressive medications early. In addition to aiding clinical decisions, my work will drive objective methods of clinical trial inclusion and data collection, improving the efficacy of randomised controlled trial (RCT) design in sarcoidosis.

Why is this research important in terms of improving the health and wellbeing of the public and to patients and health and care services?

This research will have a beneficial impact across a range of stakeholders:

Patients and carers

There is an enormous need to improve the quality of life of patients with sarcoidosis. To explore the impact of the diagnosis on patient well-being, I conducted nine interviews with patient representatives. These highlighted the huge impact of sarcoidosis on overall health. Patients stated that their symptoms were often debilitating and that treatments were unsatisfactory, sometimes resulting in intolerable side effects (two patients developed severe mental health side effects secondary to high-dose steroids, including suicidality). The diagnosis of sarcoidosis was often associated with significant anxiety due to the uncertainty surrounding the diagnosis, treatments and prognosis. Patients felt that research into sarcoidosis was neglected and were enthusiastic to discuss research priorities.

By identifying risk factors for severe disease, I hope to characterise a cohort of patients who should be initiated onto second-line immunosuppressive medications earlier. This would reduce steroid-associated side effects, improving quality of life. Patients at risk of severe disease could be recruited into clinical trials, driving the development of novel and effective treatments.

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Multi-disciplinary medical specialities

Sarcoidosis is managed by many medical specialities, including rheumatology, respiratory, cardiology, neurology, ophthalmology and dermatology teams. Interviews with patient representatives highlighted the disjointed care that patients with sarcoidosis often receive and the lack of interdisciplinary work in some cases.

My research will provide information on how different patterns of organ involvement predict outcomes and this will guide us on how care should be joined up. My research will be relevant to guideline developers on the standards of care for multi-system sarcoidosis. I will work with relevant societies, e.g. the British Thoracic Society, the British Society of Rheumatology and the European Society of Cardiology, to disseminate my work and ensure that it translates to improved patient outcomes.

Impacts on wider healthcare provision

By clarifying which patients will benefit from second-line immunosuppression, my research has the potential to reduce steroid-related side effects, improve resource allocation, improve patient outcomes, reduce disability and reduce costs for commissioners. As an example, our data estimates that 1.5% of patients with sarcoidosis go on to require home oxygen therapy. This is an expensive therapy and the total annual cost for home oxygen in the UK is approximately £120 million[8]. Even small reductions in the number who require this treatment would significantly benefit healthcare resource utilisation.

Identification of demographic predictors of severe sarcoidosis should allow appropriate targeting of healthcare resources to those who are most in need. In the longer term, through the identification and treatment of patients with an aggressive disease phenotype, I aim to reduce the burden of emergency department admissions and hospitalisation of these patients. This will ease pressures on the already-stretched NHS services.

Review of the existing evidence - how does the existing literature support this proposal?

Sarcoidosis is associated with high mortality rates. About 5% of patients with sarcoidosis die from their disease, and observational studies indicate that mortality rates are doubled compared to matched controls[9, 10,11]. Currently, we lack methods to predict which patients will die from sarcoidosis.

Sarcoidosis is an under-researched condition and disproportionately affects groups that are under-represented in research studies, including people of Black ethnicity and those with socioeconomic deprivation[12,13,14,15]. The lifetime risk for sarcoidosis is three times higher in African-American than in Caucasian Americans (0.85% versus 2.4%)[13]. Furthermore, Afro-Caribbean ethnicity has been associated with organ-threatening involvement and a poor prognosis[16,17]. These groups are under-represented in research[12]. I will address this throughout my research.

The prevalence, incidence and mortality rates from sarcoidosis are increasing over time[10,11]. Our preliminary data, using Clinical Practice Research Datalink (CPRD), demonstrates a four-fold increase in the prevalence of sarcoidosis over the last 20 years (Figure 1). There were an estimated 3,450 new diagnoses in England in 2022 (Figure 2).

Newer imaging modalities are increasing our understanding of sarcoidosis and organ-threatening disease. 18F-FDG PET-CT is a functional imaging modality that is used in the diagnosis and monitoring of patients with sarcoidosis (Figure 3). Despite its commonplace use, we do not have validated methods to quantify disease severity on PET-CT[18].

The optimal treatment of sarcoidosis is uncertain. Trials in sarcoidosis are limited by disease rarity, heterogeneity and a lack of standardised trial outcome measures[19]. Current guidelines are based on expert opinion and small uncontrolled trials[9,20,21]. Corticosteroids are the first-line treatment option but are associated with significant toxicity; observational studies link an increased dose of corticosteroids with an increased number of non-sarcoidosis-related emergency department visits, recurrent flares and higher mortality rates[22,23,24]. Steroid-sparing agents are effective treatments but are underutilised, especially outside of tertiary units[6].

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There is an unmet need for better treatments in the management of sarcoidosis. Our systematic review highlighted the lack of high-quality RCTs investigating the efficacy of second-line immunosuppressive agents in sarcoidosis. As newer treatment options are emerging, there is a need to identify patients with severe disease who are most likely to benefit from them.

My research aims to characterise risk factors for severe sarcoidosis. This work will identify patients who would benefit most from second-line immunosuppressive therapies.

What is the research question/aims and objectives?

Research question:

Which patients with sarcoidosis are most at risk of severe disease?

Key aims and objectives

My key study phases, aims and objectives are:

Phase 1: Characterise the clinical and demographic predictors of adverse outcomes in patients with sarcoidosis (0-12 months).

Phase 2: Develop a deep-learning model capable of localising and quantifying disease activity on 18F-FDG PET-CT (6-18 months).

Phase 3: Correlate activity on PET-CT with contemporary disease activity and disease outcome (12-36 months)

Project Plan

Phase 1: Clinical and demographic predictors for poor outcomes (0-12 months)

Overview
Phase 1 is a nationwide cohort study comparing the rate of severe disease outcomes between patients with and without predictors of interest.

Background
In American and Scandinavian populations, risk factors for poor prognosis in sarcoidosis include socio-economic deprivation, obesity and Black ethnicity[7,16,17]. These have not been studied in a UK population, limiting our ability to

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target healthcare resources and minimise inequalities in healthcare. This phase will investigate, characterise and quantify the risk factors for severe disease on a patient and population-based level. This will enable improved identification of patients at risk of severe disease, improving patient management and healthcare policy planning.

Data Source
I will undertake analyses of sarcoidosis epidemiology using the OpenSAFELY Trusted Research Platform, a secure nationally representative cohort of over 58 million patients, registered with general practices (GPs) in England that use TPP and EMIS electronic health record software. This large dataset is well-suited for epidemiological analyses on rare conditions, such as sarcoidosis, by increasing the likelihood of capturing a sufficient number of individuals with the disease and enhancing the statistical power of the analyses. The use of OpenSAFELY carries advantages over CPRD, a database that contains coded and anonymised data from GP practices, which includes data from only 20% of the English population.

When using OpenSAFELY, the researcher develops analytical codes on dummy data. The analytic code is sent to the live data environment and run securely on real patient data. The summary results are then reviewed by the researcher. By blocking users from directly viewing the patient's raw data, privacy risks are minimised and data safety is optimised. This has gained the trust of patient groups and is becoming the gold standard of epidemiological data analysis. Another major advantage the use of OpenSAFELY platform is the pre-existing linkage to the Office for National Statistics (ONS), Hospital Episode Statistics (HES) and Index of Multiple Deprivation (IMD).

Methods
The following definitions will be used throughout this phase of the project:

Index diagnosis of sarcoidosis – A first-ever diagnostic code of sarcoidosis between 2004 and 2024, in people aged 18 and above, permanently registered with a GP in England that uses TPP and EMIS electronic health record software.

Severe disease - Disease resulting in hospitalisation, end-organ damage or mortality. Patients will also be considered to have severe disease if they require treatment escalation or a cumulative lifetime prednisolone dose greater than 500mg.

Predictors of severe disease - Demographic and clinical factors that may predict a severe disease outcome.

Outcomes of severe disease:

Outcomes are defined as events that occur at or after the time of diagnosis of sarcoidosis. They will include; hospitalisation and end-organ damage, treatment escalation, corticosteroid burden and mortality.

Hospitalisation and end-organ damage

All-cause and sarcoidosis-related hospitalisation rates will be calculated using OpenSAFELY links to HES. Rates will be expressed in crude and age-standardised forms.

Examples of markers of end-organ damage, stratified by organ system, are listed in Table 1. As part of the project, I will develop and validate a full set of codes using the website OpenCodelist.

Treatment escalation

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Escalation to second or third-line immunosuppressive therapies will be assessed through a recorded prescription for hydroxychloroquine, methotrexate, mycophenolate, azathioprine or leflunomide. The prescription for adalimumab or infliximab will be measured using the planned link between OpenSAFELY and the targeted therapies prescribing data.

Corticosteroid burden

Cumulative lifetime corticosteroid doses greater than 500mg have been associated with adverse outcomes in large nationwide studies of patients with asthma[25,26]. Cumulative corticosteroid doses will be measured using coding of GP prescriptions for corticosteroids.

Mortality

All-cause mortality rates will be calculated using OpenSAFELY links to ONS Death Registration Data.

Mortality rates will be expressed as age-adjusted standardised Mortality Ratios, using indirect standardisation, according to England mortality data using UK-based life tables as the reference data. Mortality rates will be compared between groups with different predictors of interest.

Statistical analysis plan

Analyses will be performed to identify factors/predictors associated with the above outcomes including age, gender, ethnicity, geographical region, socio-economic deprivation, organ involvement and comorbidities. Justifications for their inclusion are outlined in Table 2.

A competing risk survival model will be performed to estimate the cumulative incidence of each outcome at 1, 2 and 5 years. The start date will be the date of the first recorded diagnosis of sarcoidosis. The end date will be the time of the outcome of interest. Multiple models will be performed, using the following outcomes of interest:

- Time to death
- Time to end-organ damage, including sub-analyses on organ systems
- Time to treatment with second or third-line immunosuppressive medications
- Time to treatment with a cumulative corticosteroid dose of more than 500mg.

Sample size calculation

I have a fixed population for my study. From my preliminary analyses of data in CPRD, we estimate that 18465 patients were diagnosed with sarcoidosis between 2003 and 2023. CPRD includes GP data from 42 million patients including many historical patients, this compares to 58 million patients that are currently registered in OpenSAFELY. Therefore, I estimate that there will be 25750 patients diagnosed with sarcoidosis in OpenSAFELY between 2004 and 2024.

Using a conservative estimate of 20,000 new diagnoses, a predictive variable that is present in just 10% of the population but explains an additional 10% increase in the risk of progression to severe disease, gives 93% power at an alpha of 0.01. This provides confidence that I will be able to detect predictors that are of clinical relevance.

Phase 2 – Quantifying disease pattern and activity on PET-CT using deep-learning techniques (6-18 months)

Overview

Pre-submission

Phase 2 will involve the development of a novel AI model to quantify disease burden on 18F-FDG PET-CT scans.

Background

PET-CT in sarcoidosis

18F-FDG PET-CT is a functional imaging modality that is used in the diagnosis and monitoring of patients with sarcoidosis. It has a high sensitivity to detect active sarcoidosis, with estimates ranging between 89% to 100% across observational studies[18].

PET-CT carries significant advantages over non-functional imaging modalities. It can differentiate active inflammation from "burnt-out" disease. PET-CT can also quantify disease activity by providing objective measures of 18F-FDG uptake using the Standardised Uptake Values (SUV). Maximum SUV (SUVmax) correlates with disease activity, including patient-reported symptoms[27], FVC[18] and cardiac events[28,29.]

Benefits of AI

AI is being used increasingly for automating time-consuming medical imaging tasks and reducing inter-observer variability. Convolutional neural network (CNN)-based approaches are a branch of deep learning that are finding application in the analysis of medical images. CNN approaches are highly effective in imaging tasks such as recognition, object detection and segmentation and studies suggest that they can reach near-human performance in well-defined tasks[30].

CNNs have proven highly effective in classifying uptake on PET-CT in cancer settings[30]. Studies have used CNNs to localise, classify and quantify foci of malignant disease[31]. In studies in lymphoma patients, CNNs have been trained to calculate metabolic tumour volume, a value that correlates with progression-free survival[31].

Methods

Development of the CNN

A CNN model will be trained to provide a measure of total organ 18F-FDG uptake based on a 3D image. Uptake in the heart and lungs will be measured as the involvement of these organs is the major cause of death in sarcoidosis[32]. Depending on the algorithm's performance, assessed using test data, measurement of uptake in other organs will be considered.

PET-CT foci of interest will be detected using segmentation with a fixed thresholding algorithm (e.g. percentage of local SUVmax or relative uptake compared to the blood pool).

The CNN algorithm will be trained using a historical cohort of PET-CTs performed at KCH in patients with biopsy-confirmed sarcoidosis. Normal PET-CTs will be used as control images in the validation process. All scans will be identified from radiology records and anonymised at study inclusion.

Manual segmentation and annotation

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Focal areas of increased 18F-FDG uptake will be manually delineated by trained radiologists, using a volume of interest tool. Identified foci will be as classified as consistent or inconsistent with active sarcoidosis. For each focus of increased 18F-FDG uptake, the classification and anatomical location (i.e. organ) will be recorded. The expert annotations will allow training of the CNN and testing of sensitivity and specificity for correct classification as consistent or inconsistent with active sarcoidosis.

CNN training and validation

PET-CTs will be randomly split into three independent subsets; validation (20%), training (60%) and testing (20%). A validation set will be used to select model hyperparameters and determine the best-trained CNN. The testing subset will be used to assess the final performance of the model[33].

Sample Size Calculation

At KCH, 10-20 PET-CTs are performed in patients with sarcoidosis every month. Using a conservative estimate of 10 PET-CTs per month over the last five years, 600 PET-CT scans will be available to test, train and validate the algorithm.

Sample size calculations are not routinely performed during the development of deep-learning algorithms. The inclusion of a higher number of scans is usually associated with improved CNN performance. Sample size will be limited by the feasibility and time required for manual delineation of FDG activity. We aim to include 300-500 scans in the development of the algorithm and believe that this is feasible. This falls in line with similar studies performed in cancer patients[30,34.]

Statistical analysis plan

The performance of the automated classification and localisation of 18F-FDG foci will be evaluated by using sensitivity, specificity and receiver operating characteristic (ROC) curves.

If the performance of the segmentation CNN and the classification network is deemed acceptable, defined as greater than 90% sensitive and specific, I will design a deep learning model capable of quantifying disease activity in the heart and the lungs. Depending on the performance of the algorithm, uptake in the liver, kidneys, eyes and lymph nodes may also be measured. A total body disease score will be generated by combining uptake in the organs of interest.

Phase 3 – Correlating disease activity on PET-CT to disease outcome (12-36 months)

Overview

Phase 3 is an observational prospective study of patients with sarcoidosis with PET-CT performed at baseline. Using the deep learning model developed in phase 2, I will correlate imaging findings to disease activity at baseline and one year later.

Background

Pre-submission

Although SUVmax has been shown to correlate with sarcoidosis disease activity, this is poorly characterised and prospective studies are lacking[32]. SUVmax is a detail-lacking measurement that does not describe the total quantity or pattern of uptake. The correlation between objective measurements of PET-CT activity and clinical outcomes could standardise PET-CT interpretation. This has the potential to standardise clinical trial design and clinical treatment decisions.

Methods

I will perform an observational prospective study of participants with biopsy-confirmed sarcoidosis with PET-CT performed at baseline. Clinical outcome markers will be measured at 0- and 12 months post-recruitment.

Participant recruitment

Inclusion criteria

- Biopsy confirmed sarcoidosis.
- Plan for PET-CT under routine NHS care, ensuring no additional radiation exposure on recruitment into this trial.

Exclusion criteria and justification

- Patients with localised neurosarcoidosis will be excluded from the study as the results will not be generalisable, due to the limited access to brain-PET-CT imaging.
- Pregnancy or post-partum, due to the harmful risks of radiation during this period.

Recruitment and consent process

Participants will be recruited from KCH. Participants will be identified from sarcoidosis MDTs and be recruited during routine clinic appointments. Recruitment will adhere to Good Clinical Practice guidelines.

Type and content of participant information materials

Before recruitment, all participants will receive Patient Information Leaflets (PILs). PILs will be co-written with patient representatives. They will outline what the study involves, including possible risks and what happens after the study finishes. PILs will explain that recruitment is optional and that consent can be withdrawn at any point.

Equality, Diversity and Inclusion for Study Participants

Participants will be recruited from KCH which serves an ethnically diverse population. Demographic data will be collected for all patients, including data on protected characteristics. I have created an Equality Impact Assessment to identify and address factors that may limit the recruitment of underserved populations into the study. To ensure adequate recruitment of participants from Black Asian and Minority Ethnic (BAME) backgrounds, I follow the NIHR Toolkit for increasing the participation of BAME groups in healthcare and social research. I have also approached the Centre for Ethnic Health Research to provide support in ensuring inclusivity.

Clinical Data

Clinical data will be collected at baseline and 12 months post-PET-CT. Data will include symptom burden, end-organ damage, hospitalisation and death and is summarised in Table 3.

Study participant support

At the time of recruitment, all participants will receive contact information for the study team (email address and telephone number). Patient representatives state that study acceptability is high, therefore, we envisage a low

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drop-out rate. If participants drop out between the baseline and 12-month visits, their data will be excluded from the predictive component of the analysis.

Lay summaries of trial results will be provided to all study participants following the completion of the trial.

Sample size calculation

A target sample size of 100 allows for 80% power to detect correlations (or standardised regression coefficients) of r>.28 at the 5% significance level for organ-specific FDG uptake with disease activity. Furthermore, based on Green's rule of thumb regression models will contain no more than 6 candidate predictors[35].

Statistical analysis

Organ-specific FDG uptake will be correlated with disease activity (table 3) at baseline. I will consider including measures of combined organ uptake, depending on the performance of the algorithm.

Regression models will be used to examine the association between organ-specific FDG-uptake at baseline and disease activity at 12 months.

A low mortality rate is expected during the period of study follow-up. Therefore, mortality will be reported descriptively. Depending on mortality and hospitalisation rates, survival analysis will be considered.

Stakeholder meetings:

Stakeholder meetings will take place in all three phases of the project. Patient representatives, Sarcoidosis UK employees and colleagues from respiratory, cardiology and radiology specialties will be invited. Stakeholders will be asked to provide feedback on study design and dissemination of study findings. Two meetings will take place in person and one meeting will take place online.

Dissemination, outputs and anticipated impact

Academic outputs:

I will publish findings from each phase of my research plan in peer-reviewed journals with

high impact factors. I anticipate three key manuscripts; the first, centred on my epidemiological analyses of sarcoidosis (Phase 1); the second, detailing the use of deep-learning techniques of PET imaging to quantify disease activity in sarcoidosis (Phase 2); the third, describing the relationship between sarcoidosis activity on PET-CT and contemporary disease activity and patient outcome (Phase 3). I will target journals with wide audiences, ensuring health professionals from multiple clinical specialties can access my research outcomes.

I will present my findings at rheumatology and thoracic conferences. I will use the links that I have built with cardiology and respiratory colleagues to disseminate findings to a broad range of medical audiences.

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Patient outputs:

I have engaged the support of the Sarcoidosis UK Charity as a collaborator to support dissemination strategies to patients and the wider population. My PPI panel will lead on dissemination strategies that will include:

1. Plain English Summaries to accompany all publications
2. Social media summaries of outputs (including the Facebook Sarcoidosis UK page)
3. Public engagement events (via BSR, EULAR, RCP and Sarcoidosis UK patient council)
4. Newletters via the Sarcoidosis UK charity

Healthcare and societal outputs:

My research will inform the design of clinical trials of immunosuppressive medications for sarcoidosis on a global scale. My research will allow non-expert clinicians to identify patients at risk of life-threatening sarcoidosis, who should be considered for referral to specialist centres and treatment with second-line immunosuppressive medications. This will improve the care of patients with sarcoidosis, reducing hospitalisations and death.

Impact statement:

My goal is to produce robust and objective methods that are capable of accurately predicting sarcoidosis patients at risk of severe disease. By identifying high-risk patients, I hope to identify:

- Patients who should be referred to specialist tertiary centres for treatment.
- Patients who will benefit from second-line immunosuppressive therapies, reducing corticosteroid burden and the risk of flare when steroids are tapered.
- Patients who should be recruited into clinical trials of novel therapies.

By achieving the above, my work will reduce symptom burden and the risk of end-organ damage and mortality. My work will also drive objective methods of RCT design and inclusion criteria, improving the efficiency of research, which is important for research in a rare condition.

Following my fellowship, I will apply for an NIHR Clinical Lectureship and use my data to inform a nationwide study of novel treatments for sarcoidosis. In the longer term, the skills that I develop will enable me to reach my goal of becoming a leading clinical academic in the field of sarcoidosis.

Project management

I have created a research timetable (a Gantt chart), which I will follow during my fellowship.

I will register for a PhD and be supervised according to the KCL PhD supervision structure. This provides a robust framework for official assessments every 6 months and a formal external peer review at one year.

King's College London has an excellent academic support structure to facilitate all aspects of my research plan, including the epidemiological, deep-learning and prospective trial components. The Centre for Rheumatic Disease is a EULAR Centre of Excellence for rheumatology research. KCH treats a large cohort of sarcoidosis patients, with monthly multi-disciplinary sarcoidosis meetings. The hospital serves a multicultural and diverse population, enabling study recruitment with participants from many ethnicities. The King's Clinical Trials Unit runs high-quality clinical trials, including three studies investigating novel therapies in the management of sarcoidosis.

Professor Galloway is a rheumatologist and methodologist, with extensive experience in epidemiological, statistical and clinical trials-based research and the supervision of PhD students. Within Dr Galloway's research group, I will have access to a network of qualitative researchers, health psychologists, epidemiologists and statisticians to support me during my Fellowship.

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Dr Katie Bechman is a rheumatologist and methodologist. She has a wealth of experience in epidemiological and statistical analyses using large health datasets. Dr Bechman and Professor Galloway have an interest in sarcoidosis and collaborate closely with cardiology, respiratory and radiology colleagues during weekly sarcoidosis MDTs.

Professor Andrew Reader is a Physicist and Professor in Imaging Sciences. He has extensive experience in the use of deep-learning techniques in PET-CT imaging. He has a wealth of experience in supervising doctoral students through the completion of PhDs. He bases his research in the School of Biomedical Engineering and Imaging Sciences, a collaborative centre with plenty of expertise to support the imaging aspects of my project.

During my Fellowship, I will hold monthly meetings with my supervisors and biannual meetings with my collaborators, to present progress reports. I will work with the KCL finance office to track my budget. I will maintain a risk register throughout my PhD which will be regularly reviewed with my supervising team.

Ethics

Phase 1

I will undertake epidemiological analyses using OpenSAFELY. No individual data will be visible or transferred to me. OpenSAFELY adheres to the principles of General Data Protection Regulation (GPRD).

Phase 2

Ethics approval will be obtained for Phase 2 and 3. I will anonymise PET-CTs at study inclusion and remove all patient-identifiable information.

Phase 3

I will only recruit patients with PET-CT performed during routine clinical care. This will ensure that patients are not exposed to unnecessary radiation. I have discussed this strategy with patients at the focus group, who agree with this strategy.
I will apply for ethics approval for Phase 2 and Phase 3 of this project. Before recruitment into this study, written informed consent will be obtained. As clinical data will be limited to observational data, I do not envisage any ethical issues arising from this.

The handling of all data in this project will be performed in compliance with the Caldicott principles and GPRD.

Success criteria

Phase 1:

- Completion of an epidemiology and statistics course
- Presentation of my findings at a national conference
- Publication of a manuscript detailing the demographic and epidemiological risk factors for severe disease in sarcoidosis.

Phase 2

- Completion of a course in coding, using Python, and in deep learning.
- Presentation of my findings at a national conference
- Publication of a manuscript outlining the deep learning techniques used to quantify and localise disease activity on PET-CT.

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Phase 3:

- Presentation of my findings at an international conference
- Publication of a manuscript detailing the prognostic value of PET-CT in sarcoidosis

I have identified the following risks to delivering this research, with actions to mitigate these.

Phase 1

- Delays in access to OpenSAFELY
  - I will engage the support of Brian Mackenna, the direct of NHS Service Analytics, and submit my data request 6 months before commencing my fellowship.
  - I have costed for the use of CPRD data as a contingency plan as our group already has access to this. CPRD has a smaller dataset and therefore the use of OpenSAFELY is preferred.

Phase 2

- Inability to develop a sufficiently accurate segmentation CNN and classification network.
  - If I am unable to develop a tool of sufficient sensitivity and specificity, I will work with a radiology colleague to manually quantify FDG-avidity. I have costed in for the time of a radiologist throughout Phase 2 and 3 of my study.
- Time-consuming nature of manual segmentation and annotation of PET-CT scans
  - After labelling enough scans, I will train a U-net (a CNN tool) to automatically label organ-specific activity. If the performance of the U-Net is adequate, this will significantly reduce the time needed for segmentation and annotation.
-

Phase 3

- Inability to produce a deep learning model capable of measuring PET-CT activity
  - If Phase 2 fails, I will measure PET-CT activity manually, with the help of a radiology colleague, who has been costed for both Phase 2 and 3.
- Delays in obtaining ethical approval for the observational trial.
  - I will submit my ethics application early in the course of my Fellowship to allow sufficient time to address feedback.
  - I will work with stakeholders (including patients and public representatives) in the design of my study and ethics submission, increasing the likelihood of a successful outcome.
- Difficulties in patient recruitment and loss of patients to follow-up
  - The KCH radiology team perform 10-20 PET-CT scans in patients with sarcoidosis every month. Therefore, I do not foresee an issue with the availability of PET-CT images to train and validate the algorithm.
  - I will work closely with patient representatives to ensure that the trial is acceptable to patients, maximising recruitment and minimising drop-out.
  - I have already gained support from colleagues in respiratory and cardiology medicine. I will collaborate with colleagues from other specialities to raise awareness of the trial and increase recruitment.

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7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

I engaged the support of the charity Sarcoidosis UK to identify patient representatives to discuss research ideas. Participants ranged in age, gender and ethnicity, ensuring that different perspectives were provided.

I organised a focus group of patients with multi-system sarcoidosis, highlighting the enormous impact that sarcoidosis can wreak on a patient's life:

- "I'm basically more or less housebound. I use oxygen when I move."
- "It [sarcoidosis] can absolutely be devastating"

The focus group highlighted patient priorities for research, summarised below.

Patient priority	Patient Quote	Impact on proposal design
Increased treatment options	"I went through steroids, hydroxychloroquine and methotrexate and after that there wasn't much left in the cupboard"	My research proposal will identify the group of patients who will benefit from second-line immunosuppressive medications. My research will drive the standardisation of RCT design and the development of new therapies.
Inclusion of a range of disease activity measures in study design	"The symptom list is kind of almost like endless when it comes to sarc [sarcoidosis]. So it's tick ticks, so many different boxes, because everyone experiences different things and a lot of different things at the same time. So then you become, you kind of feel almost embarrassed to have to tell your specialist."	During Phase 2, I will measure disease outcome using multiple validated disease activity scores, considering symptoms across all relevant body systems.
Improved multi-disciplinary working	"They don't really look at it together, you know, it's very much in isolation"	Throughout my research proposal, I plan to collaborate with specialists from relevant medical specialities. My research will drive improved holistic management of sarcoidosis.
Improved awareness of sarcoidosis	"I think there's a lot of misunderstanding and even misinformation even amongst medical professionals"	I will disseminate my research findings to a range of medical specialities through medical conferences and publications.
A sense of feeling heard	"I got told to see a psychologist because of all the symptoms that it was just soul destroying and literally just cried and cried because you feel so unheard."	Patients will be involved in all aspects of my project, allowing their voices to be heard.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

Prior to starting my fellowship, I will register for PPI training sessions through the NIHR University College Hospital Biomedical Research Centre team.

I have already identified six patients with multi-system sarcoidosis who will form a patient advisory group. Early in my research, I will invite additional stakeholders, including carers and members of the Sarcoidosis UK Charity, to

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join the group. This process will be supported by King's College London Centre for Rheumatic Disease. Participants will receive Public Involvement Training organised through the Patient Experience Research Centre at Imperial College London. This course is free for patient representatives.

I will meet with my patient panel bi-annually throughout my project. Meetings will occur online and in person. Throughout my project, I will follow the INVOLVE guidance for the payment and reimbursement of expenses. Participant involvement will be acknowledged through publication authorship.

I will work with a patient representative of BAME background, to identify factors that may limit recruitment of ethnic minorities into the observational trial. This patient will be offered an honorary research contract and has agreed to support my research proposal throughout.

During Phase 1, participants will be invited to discuss predictors and outcomes of interest and will aid design of the study protocol. In Phase 2, I will invite patient representatives to join meetings with Physicists and Radiologists to design the deep learning model to measure disease activity on PET-CT.

The patient advisory group will join the stakeholder meeting to develop the observational trial including measures of disease severity, including the inclusion of different symptom questionnaires (Phase 3). We will work together to ensure that the study is acceptable to patients. The patient advisory group will support the writing of patient information leaflets and lay summaries.

To disseminate my research findings, I will work with my panel to:

- Produce lay summaries of all research outputs.
- Engage with the charity Sarcoidosis UK to publish my findings on their website and their Facebook page (identified by patients as a source of trusted information).
- Arrange local public engagement events, including research days and science festivals.

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Not applicable

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8. Training & Development and Research Support

Proposed training and development programme

1. **Academic training and courses**

To advance my knowledge and skills in epidemiological analysis, I will complete the 'Advanced Course in Epidemiological Analysis' hosted by the London School of Hygiene and Tropical Medicine. This two-week course covers a range of epidemiological methodologies in Stata, which will be extremely useful for Phase 1 of my project and my future research career.

As a KCL student, I will take advantage of the abundance of learning opportunities. To develop skills in programming, using Python, I will complete the KCL module "Introduction to Programming". To further my understanding of deep learning, I will complete the "KCL Masters modules on machine learning and advanced machine learning", focusing on the modules in medical imaging.

I will also attend the international "Summer School on Deep Learning for Medical Imaging". This one-week course covers the basic and advanced aspects of deep learning, which will prove extremely useful for Phase 2 of my study.

I will attend the British Medical Association (BMA) Foundation and Advanced Level Leadership Programmes. These are free to BMA members and cover topics including leadership, personal awareness, resolving conflict and facilitating collaborative decision making.

1. **Conferences and research placement**

I will attend conferences in each year of my Fellowship, allowing me to present my research findings and network with rheumatology and relevant medical specialists. These will include British Society for Rheumatology Annual Conference (Year 1), the British Thoracic Society (Year 2) and the American College of Rheumatology Annual Meeting (Year 3).

Through my collaboration with Professor Matthew Baker, I will complete a three-week placement at Stanford University Medical Centre, within the Sarcoidosis Programme. This is run by a multidisciplinary team, including specialists in rheumatology, cardiology and respiratory medicine. Through this placement, I will learn from world-leading academic clinicians and build strong collaborative links that will be extremely useful for my ongoing career. From a clinical perspective, I will gain clinical knowledge regarding the management of sarcoidosis from the Sarcoidosis Centre of Excellence, recognised for exceptional patient care. Through my placement, I will gain exposure to a range of research projects in sarcoidosis, ranging from bench-to-bedside translational research projects to clinical trial projects.

I have costed in courses, conferences and a research placement in my application.

2. **Formal study**

I will enrol in a PhD programme in the School of Immunology and Microbial Sciences at King's College London. As part of the Doctoral Training Programme, I will select a range of courses to support my research proposal and my development as an academic clinician, including: Introduction to Statistics for Researchers, Managing Your Research Data; Leadership Skills for Research Students; Writing up the Thesis and Preparing for the Viva. I will also attend courses on using "Stata".

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3. **Clinical training and development**

Throughout this fellowship, I will attend one NHS clinic per week and one multi-disciplinary meeting per month at KCH, under the supervision of Professor Galloway. I will rotate through a range of speciality clinics, with a particular focus on sarcoidosis clinics to complement my research and build upon my clinic knowledge of this condition. I will also perform one evening of rheumatology on-call duties per week, to maintain my knowledge of rheumatology emergencies and inpatient management.

During the fellowship, I will maintain an up-to-date clinical portfolio, including a yearly personal development plan and regular work-based assessments. To do this, I will contribute to my NHS e-portfolio regularly and have bi-yearly meetings with a clinical supervisor. I will attend Annual Reviews of Competence Progression, ensuring that I meet and maintain relevant clinical competencies.

RESEARCH SUPPORT / SUPERVISOR 1

Name	Professor James Galloway

Institution	King's College London

Position	Professor of Rheumatology

Description of support provided
Professor Galloway is a Professor at King's College London and a consultant rheumatologist at King's College Hospital. His extensive experience in epidemiology, statistics and clinical trial design makes him highly qualified to supervise my proposed research project. Through Professor Galloway's research group, I will have access to a network of epidemiologists and statisticians to support me during my Fellowship. Professor Galloway has successfully supervised nine PhD students to completion. Additionally, he is the MBBS Chair of Assessment Board Programme Director and a Faculty Tutor for the PGCert in Applied Research Methods. Professor Galloway has extensive experience using the OpenSAFELY platform to publish studies in high-impact journals. He has authored over 250 papers in peer-reviewed journals, with a Scopus H-index of 43. Professor Galloway has a clinical and research interest in sarcoidosis, providing rheumatology expertise to the sarcoidosis multi-disciplinary meetings at King’s College Hospital. He has an excellent working relationship with Professor Baker, Professor Birring, Dr Sado and Dr Bechman, providing a good multi-disciplinary team to support my application.

RESEARCH SUPPORT / SUPERVISOR 2

Name	Professor Andrew Reader

Institution	King's College London

Position	Professor of Imaging Sciences

Description of support provided

Pre-submission

Professor Reader is a Physicist and Professor in Imaging Sciences. He has extensive experience in deep learning techniques in PET-CT imaging. He has a wealth of experience in supervising doctoral students through the completion of PhDs. He bases his research at the School of Biomedical Engineering and Imaging Sciences, a collaborative centre with plenty of expertise to support the imaging aspects of my project.

Professor Reader has a wealth of experience in medical image reconstruction and analysis in PET-CT imaging. He has authored over 240 papers in peer-reviewed journals, with a Scopus H-index of 34. He is also the Senior Editor for the British Journal of Radiology | Artificial Intelligence.

RESEARCH SUPPORT / SUPERVISOR 3

Name	Dr Katie Bechman

Institution	King's College London

Position	Clinical Lecturer in Rheumatology

Description of support provided

Dr Bechman is a Clinical Lecturer at King's College London and a Rheumatology Registrar at King's College Hospital. She has extensive experience in epidemiology and statistics, making her very well qualified to supervise Phase 1 of my research project.

Dr Bechman has an impressive publication track record. She has authored over 60 papers in peer-reviewed journals, with a Scopus H-index of 20. Professor Galloway and Dr Bechman have collaborated on multiple projects and have an excellent working relationship.

Dr Bechman has a clinical interest in sarcoidosis and collaborates closely with cardiology, respiratory and radiology colleagues during monthly sarcoidosis MDTs.

Before this application, I worked with Dr Bechman and Professor Galloway on a systematic review of biological medications in sarcoidosis. We have also recently submitted an abstract to the American College of Rheumatology on the epidemiology of sarcoidosis in England and are writing a manuscript for submission for publication.

Dr Bechman and Professor Galloway have close collaborative links with the Charity Sarcoidosis UK and have written several patient information leaflets for their website.

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Collaborations

I have engaged the support of medical specialists in the fields of cardiology, respiratory and radiology in this research proposal.

Professor Surinder Birring is a Consultant Respiratory Physician and Professor of Respiratory Medicine at KCH and KCL. He has a specialist interest in sarcoidosis, leading the sarcoidosis clinic at KCH and publishing widely in this area. He is a co-author of the European Respiratory Society guidelines for the treatment of pulmonary sarcoidosis and developed the Kings Sarcoidosis Questionnaire.

Dr Marko Berovic is a radionucleotide radiologist with expertise in cardiothoracic and musculoskeletal imaging. He has extensive experience in the use of PET-CT in sarcoidosis and joins the sarcoidosis MDT at KCH monthly.

Dr Daniel Sado is a consultant cardiologist, with a specialist interest in imaging. He has a particular interest in heart failure, particularly secondary to sarcoidosis.

Professor Matthew Baker is the Clinical Chief in the Division of Immunology and Rheumatology at Stanford University and the Co-Director of the Stanford Multidisciplinary Sarcoidosis Program, which is a Centre of Excellence for the management of sarcoidosis. He has published widely on sarcoidosis.

Host Organisation Support Statement
The following statement of support must be completed by the head of your lead academic department. In order to contribute to this application, this individual must be invited to do so via the 'Participants and Signatories' section of the form.

Statement of support

The Department of Inflammation Biology is delighted to support Dr Kathryn Biddle's application for an NIHR Doctoral Fellowship. I have read her application and believe it to be novel, well-designed and impactful.

Dr Biddle committed herself to research from an early stage, achieving First-Class Honours in her intercalated degree in Neurosciences at Cambridge University. She excelled academically during medical school, earning prizes and graduating with a Distinction. She gained further research experience by completing an MSc in Immunology at the University of Oxford and then pursuing an Academic Clinical Fellowship (ACF) at St George's University. She is the first author of five PubMed-cited publications and a co-author of a further three publications. She has a wealth of experience in communicating her work, through several oral and poster presentations, at national and international conferences.

Dr Biddle has excelled throughout her clinical and academic career; I believe that she is an excellent candidate for an NIHR Doctoral Fellowship. She is dedicated, highly motivated and conscientious. Her ability to write this application whilst juggling full-time clinical duties is a testament to her hard-working and ambitious nature.

Dr Biddle's research outputs are evidence of her dedication. Despite her ACF being interrupted by the COVID-19 pandemic, she published a research manuscript, several review articles and case reports. She also authored a book introducing healthcare professionals to research topics.

Alongside her clinical duties and research commitments, Dr Biddle regularly contributes to undergraduate teaching. She provides introductory lectures on rheumatology and tutorials on inflammatory arthritis, with excellent feedback. She has also excelled in her leadership roles including the Royal College of Physicians Associate Clinical Tutor and Chief Rheumatology Registrar.

Dr Biddle developed a keen interest in sarcoidosis during her clinical training. During her involvement with our department, Dr Biddle has performed a systematic review of the efficacy of biological and targeted synthetic disease-modifying anti-rheumatic drugs in sarcoidosis. This work has been submitted for publication. She has also worked with Dr Bechman and Professor Galloway to describe the current epidemiology of sarcoidosis in the UK using data from Clinical Practice Research Datalink. To develop collaborative relationships, she is organising a Sarcoidosis Study Day which will be attended by experts from a range of relevant specialities.

KCL will provide robust training oversight for Dr Biddle. She will be based at the Weston Education Centre and have access to the School of Biomedical Engineering and Imaging Sciences. Her primary supervisor, Professor Galloway, has a wealth of experience supervising clinical PhD students and receives excellent feedback from his supervisees. The success of his mentorship is highlighted by Dr Bechman's progress from his PhD supervisee to a NIHR clinical lecturer. She will make an excellent co-supervisor for Dr Biddle's project.

Professor Galloway's group is ideally suited to support clinical epidemiology PhDs with excellent dry lab support and a large group of pre and post-doctoral students who collaborate closely. The group has a very successful

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publication record of high-impact systematic reviews and epidemiology studies (including several OpenSAFELY projects).

Dr Biddle's final co-supervisor, Professor Reader, has expertise in deep learning in medical imaging. He has extensive experience in supervising PhD students through to completion and has received wonderful feedback from his students. Through Professor Reader, Dr Biddle will have access to the School of Medical Imaging. This will provide an abundance of expertise to guide the deep learning components of her study.

Dr Biddle has established exciting collaborations with clinicians from relevant specialities, including radiology, respiratory and cardiology medicine. Professor Birring has published widely on sarcoidosis, leads the sarcoidosis clinic at King's College Hospital and oversees several clinical trials at the Clinical Trials Unit. Dr Sado provides expertise in cardiac sarcoidosis and Dr Berovic has a wealth of experience in PET-CT imaging in sarcoidosis.

I fully support Dr Biddle's plans to undertake a research placement at Stanford University which houses a Sarcoidosis Centre of Excellence. This will provide an invaluable opportunity to gain clinical and research experience in sarcoidosis and network with international colleagues.

Dr Biddle's research proposal has the potential to identify patients at risk of severe forms of sarcoidosis. This would enable patients to receive earlier treatment with second-line immunosuppressive medications, reduce corticosteroid-related side effects and improve patient quality of life. Objective methods of PET-CT interpretation have the potential to drive evidence-based clinical decisions. This study has the potential to change the way that clinical trials in sarcoidosis are designed and conducted.

In conclusion, the Department of Inflammation Biology offers its unreserved support to Dr Biddle and her Fellowship Application. The Department is fully committed to ensuring that the training, supervision and infrastructure are in place to facilitate her project. We wish her the best of luck in her application.

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9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Kathryn Biddle

Employing organisation	King's College London

Grade	ST

Starting point on scale	ST6

Increment Date	04/01/2025

Justification of Costs
I have costed my salary for 3 years at 100% whole-time equivalent. This includes my basic salary (including ST6 nodal pay point), London geographical allowance, national insurance contributions, superannuation, and apprenticeship levy (under other allowances).

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	63152.00
Geographical Weighting (£)	2162.00
Other allowances (£)	326.57
Superannuation (£)	9470.53
Nat. Insurance (£)	7757.53
Current Annual Costs (£)	**82868.63**

Staff Type: (Lead Applicant)-Dr Kathryn Biddle
Type of cost: HEI
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: I have costed my salary for 3 years at 100% whole-time equivalent. This includes my basic salary (including ST6 nodal pay point), London geographical allowance, national insurance contributions, superannuation, and apprenticeship levy (under other allowances).

	Year 1	**Year 2**	**Year 3**	**Total Cost (£)**
Basic Salary	63152	63152	63152	189456

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National Insurance	7758	7758	7758	23274
Superannuation	9471	9471	9471	28413
Geographical weighting / Other allowances (if applicable)	2489	2489	2489	7467

Staff Type: Other
Other Type: Shared Staff
Name: **Dr Marko Berovic**
Type of cost: NHS
% Full-time on this research: 10.0
Total months on this research: 24.0
Justification of Cost: I have costed a consultant radiologist as an NHS partner. He will be costed at 10% whole-time equivalent whilst the deep learning algorithm is developed.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	15846	15846	0	31692

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	198888	0	0	198888
Total NHS	0	0	31692	31692
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference Fees, Travel and Accommodation
Justification of Cost: I will utilise national and international conference forums to present my research findings,seek feedback on my preliminary results, and disseminate my final findings. I have costed for attendance at the British Society for Rheumatology annual conference in the first year: conference registration fees of £300; 3 nights of accommodation per conference (£70 per night), and travel via a super off-peak return train ticket (£50 per conference). Total costs = £560. I have costed for attendance at the British Thoracic Society Winter Meeting in the second year: conference registration fees of £440; 2 nights of accommodation per conference (£70 per night), and travel via a super off-peak return train ticket (£50 per conference). Total costs = £630. I have costed for attendance at one international meeting in the final year of my Fellowship: conference registration fees of £300 (using the American College of Rheumatology annual congress as an example); 5 nights of accommodation at £100 per night (£500 total); travel via a return flight (£800). Total costs = £1600.

	Year 1	Year 2	Year 3	Total Cost (£)
	560	630	1600	2790

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Travel costs to overseas placement
Justification of Cost: In year 3 of my research plan, I will complete a three-week placement at the University of Stanford, USA. Travel costs: £1000

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	1000	1000

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Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Accommodation and subsistence for overseas research placement
Justification of Cost: In year 3 of my research plan, I will complete a three-week placement at the University of Stanford, USA. Three weeks of accommodation at a basic hotel and food estimated at £3525.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	3525	3525

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	560	630	6125	7315

Equipment

Type of cost: HEI
VAT relief to be applied: True
Description: Computer
Price excluding VAT £: 700
Expected life time: 50
Justification of Cost: I will require a computer with sufficient processing power and memory to enable it to complete resource intensive calculations in Python and stata.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	672	0	0	672

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	672	0	0	672

Summary

	Travel (excl Conferences)	Travel (Conferences)	Equipment
Total HEI	3620	2790	672
Total NHS	0	0	0
Total Commercial	0	0	0
Total Other	0	0	0

Consumables
No costs have been added

Patient and Public Involvement

Description: Patient focus groups
Type of cost: HEI
Please describe fully: Patient focus groups
Justification of Cost: I have costed for in-person patient focus groups in Phase 1, Phase 2 and Phase 3 of my project. Online focus groups will take place between the in-person events.

	Year 1	Year 2	Year 3	Total Cost (£)
	600	0	300	900

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Description: Public engagement events
Type of cost: HEI
Please describe fully: Public engagement events
Justification of Cost: I have costed for three events to be organised with my PPI panel to enage with the public and disseminate our research findings. Patients and members of the public will be invited to attend through organisations such as Sarcoidosis UK.

	Year 1	Year 2	Year 3	Total Cost (£)
	400	400	400	1200

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	1000	400	700	2100

Other Direct Costs

Description: Access to Clinical Practice Research Datalink (CPRD)
Type of cost: HEI
Please describe fully: Access to Clinical Practice Research Datalink (CPRD)
Justification of Cost: I have costed for the use of Clinical Practice Research Datalink (CPRD) as a contingency plan in the event that I cannot gain access to OpenSAFELY. £10,000 covers my contribution to the multi-site license in place at King's College London.

	Year 1	Year 2	Year 3	Total Cost (£)
	10000	0	0	10000

Description: Stakeholder meetings
Type of cost: HEI
Please describe fully: Stakeholder meetings
Justification of Cost: I have costed for two in-person stakeholder meetings during the project. These meetings will be attended by clinicians from relevant specialties, charity representatives, patients and members of the public.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	3000	3000	6000

Description: Advanced Course in Epidemiological and Medical Statistics
Type of cost: HEI
Please describe fully: Advanced Course in Epidemiological and Medical Statistics
Justification of Cost: I will attend the "Advanced Course in Epidemiological Analysis" at the London School of Hygiene and Tropical Medicine to provide me with the skills required to perform the analyses in Phase 1 of my project.

	Year 1	Year 2	Year 3	Total Cost (£)
	2358	0	0	2358

Description: Summer school on deep learning for medical imaging
Type of cost: HEI
Please describe fully: Summer school on deep learning for medical imaging
Justification of Cost: I will attend the "Summer school on deep learning for medical imaging" online. This will provide me with the skills that I need to develop the deep learning model in Phase 2 of my study.

	Year 1	Year 2	Year 3	Total Cost (£)
	66	0	0	66

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	12424	3000	3000	18424

Training And Development

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Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: PhD Tuition Frees
Justification of Cost: I have costed for PhD tuition fees at King's College London for each year of my fellowship. The applicable King's College London PhD fees are £7360 per year.

	Year 1	Year 2	Year 3	Total Cost (£)
	7360	7361	7361	22082

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	7360	7361	7361	22082

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	101473	92853	81837	276163

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: King's College London
Justification of Cost: Estate costs, including infrastructure technician costs, have been calculated by Kings College London finance team.

	Year 1	Year 2	Year 3	Total Cost (£)
	31642	31642	31642	94926

Type of cost: Other Indirect Costs
Name of Higher Education Institute: King's College London
Justification of Cost: Other indirect costs have been calculated by King's College London finance team

	Year 1	Year 2	Year 3	Total Cost (£)
	84154	84154	84154	252462

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	115796	115796	115796	347388

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	92637	92637	92637	277911

Research Costs Requested From Funder

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	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	219599	277911	497510
Total Higher Education Institution Costs (100%)s	24872	0	24872
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	31692	0	31692
Total Research Costs Requested From Funder	276163	277911	554074

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**82142**	**82142**	**66296**	**230580**
Travel-subsistence-and-conference-fees	**560**	**630**	**5220**	**6410**
Equipment	**672**	**0**	**0**	**672**
Consumables	**0**	**0**	**0**	**0**
Patient-and-Public-Involvement	**800**	**320**	**560**	**1680**
Other-Direct-Costs	**9939**	**2400**	**2400**	**14739**
Training And Development	**7360**	**7361**	**7361**	**22082**
HEI-Indirect-costs	**92637**	**92637**	**92637**	**277911**
Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**194110**	**185490**	**174474**	**554074**

NHS Support Costs

No costs have been added

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs

No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	Yes

Is the patient/service user care being provided different from the usual standard treatment for the condition?
No

Usual Treatment Costs

No costs have been added

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Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	0
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	0

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	554074
Total NHS (Support and Excess Treatment) Costs	0
Total Funding Required	554074

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10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as a .csv file or Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - References(2).docx) is included as an appendix within this file.

(Figures / Tables - Fellowship-combined.pdf) is included as an appendix within this file.

(Research Timetable - Gantt Chart.pdf) is included as an appendix within this file.

(SoECAT Template - Funder Export_346985_SoECAT_PAROS Study.xlsx) is included as an appendix within this file.

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11. Participants and Signatory

Participants
New participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please see the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

You are required to supply the names and email addresses (if not already registered with this online application system) of each required participant. Everyone **newly** named in this section will be acting as a 'participant' to your application and will need to agree to be part of this application.

By confirming participation, participants are acknowledging their involvement and input into this application, and agreeing to be involved in the proposed fellowship. You must ensure all **new** participants are happy for your application to be submitted before submitting it on the online system.

It is permissible for a single individual to serve as a signatory and a participant, but they will need to be added separately under each heading.

Details of the individual who will be required to approve your application after submission should be entered in the 'Participants and Signatory' section. Please note that it is acceptable for a Head of Department to also act as a research support role. If this is the case, they will need to be included in this section for both roles.

Participants must confirm their participation on your application before you will be able to press the submit button. Please see the 'Application Submission Process Flow Diagram' in the guidance notes for further information.

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Doctoral Primary Supervisor

Doctoral Primary Supervisor Participant

Title	Professor
Forename(s)	James
Surname	Galloway
Position	Professor / Honorary Consultant

Supervisor

Supervisor Participant

Title	Dr
Forename(s)	Katie
Surname	Bechman
Position	Specialist trainee year 5

Supervisor Participant

Title	Professor
Forename(s)	Andrew
Surname	Reader

Position

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Dr
Forename(s)	Antoni
Surname	Wojcik
Position	

Signatory
Participants must complete their actions on your application before submission; the signatory must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

You are required to supply the name and email address (if not already registered on the ARAMIS application system) of the individuals who will be 'signing off' your application. Once their contact details have been entered below, these individuals will be invited to log in and confirm their participation in your application.

The signatory must ALSO approve the application AFTER you have selected the 'SUBMIT' option but BEFORE the application submission deadline. Please see the Submission Process Flow Diagram and Applicant Guidance Notes for further information.

The signatory approval will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatory) will be notified of this via an automated system-generated email.
NIHR will not accept any applications unless fully approved by your signatory before the 1:00 pm deadline on Thursday 18th July 2024.

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Host Organisation - Heads of Department

Head of Department or Senior Manager

Title	Dr
Forename(s)	Tanya
Surname	Shaw
Position	Senior Lecturer

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
I have no conflicts to declare

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions:
I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR Research Governance Framework.

If you are applying for an NIHR award which comes with partnership funding from another organisation (e.g. a charity or industry partner) your information (including your NIHR Application Form and subsequent progress/annual reports) may be shared with the partner organisation as part of the review process and ongoing monitoring of the award.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Pre-submission

Appendices

1. References(2).docx
2. Fellowship-combined.pdf
3. Gantt Chart.pdf
4. Funder Export_346985_SoECAT_PAROS Study.xlsx

Pre-submission

Appendix #1: References(2).docx

Pre-submission

References

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Appendix #2: Fellowship-combined.pdf

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Figures and Tables

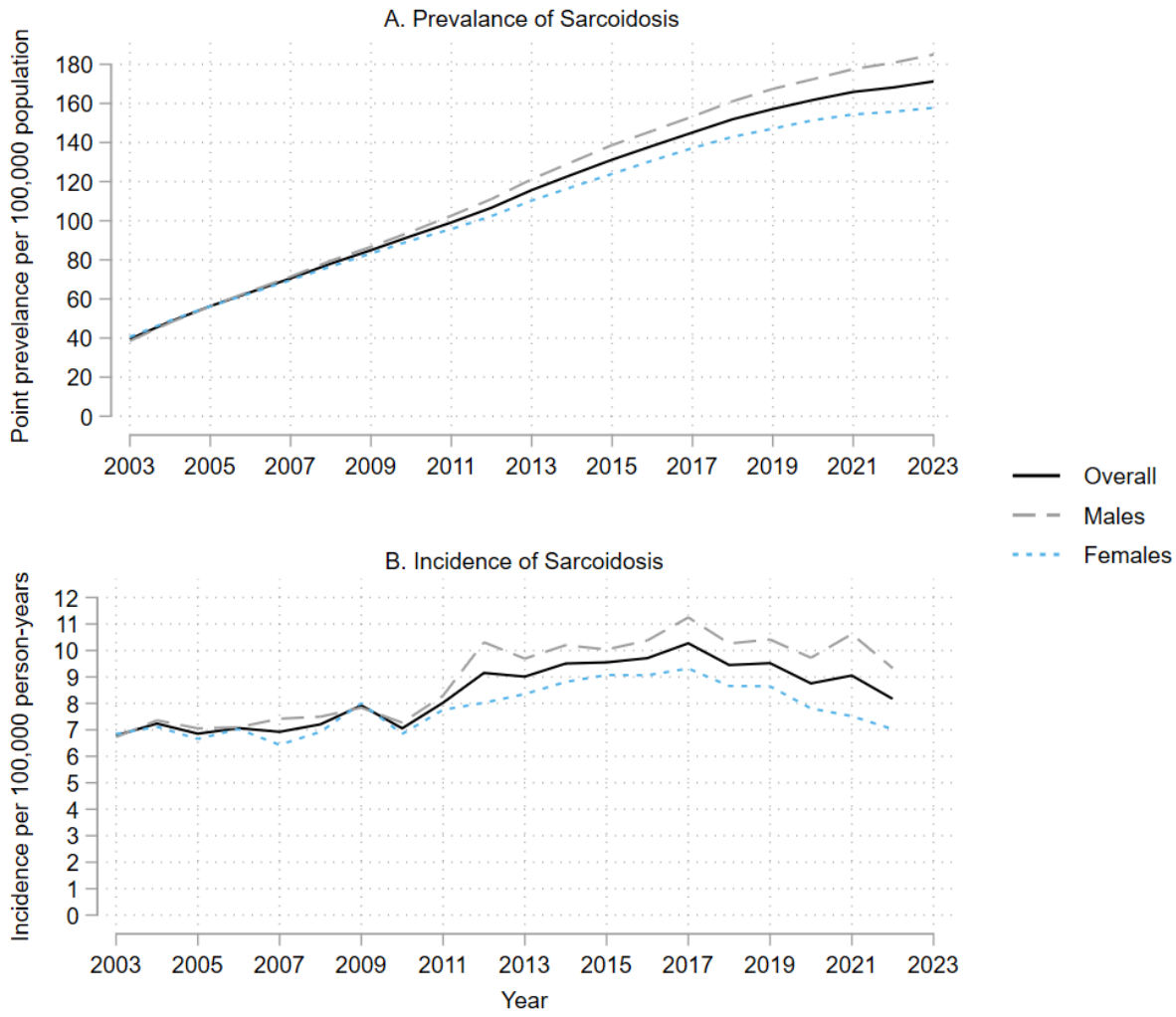

Figure 1. Our preliminary data from the Clinical Practice Research Datalink (CPRD) illustrating the change in the prevalence and incidence of sarcoidosis from 2003 to 2023.

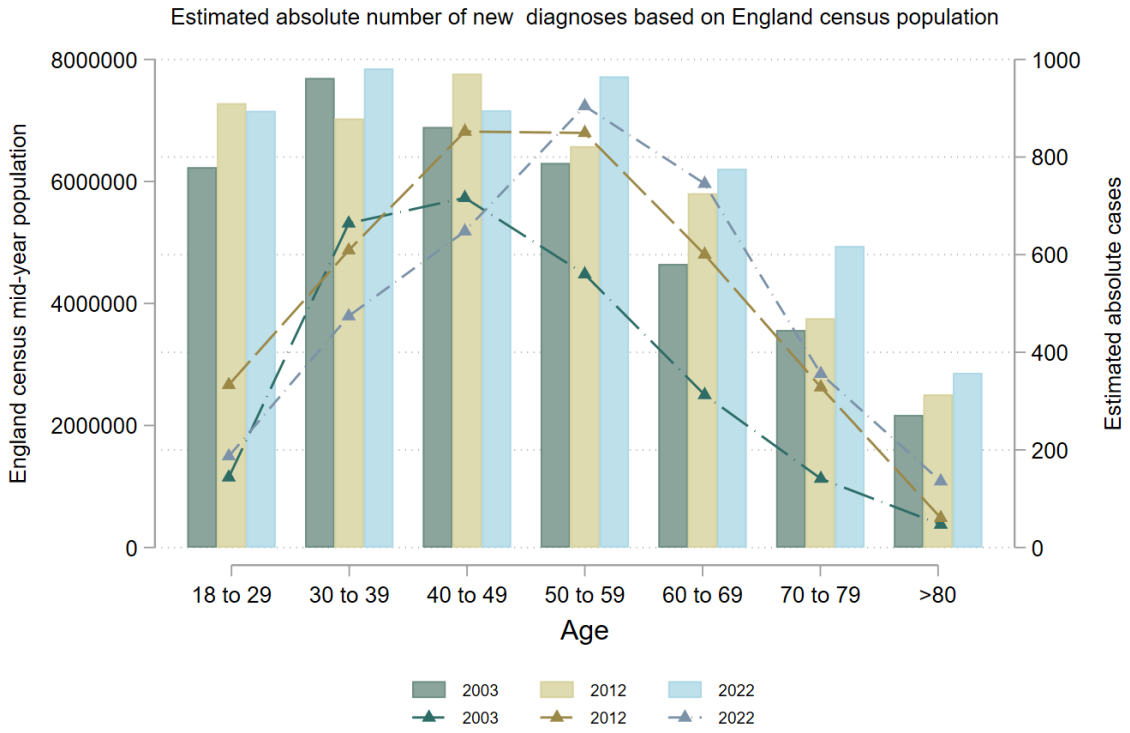

Figure 2. Our preliminary data from the Clinical Practice Research Datalink (CPRD) illustrating the estimated absolute number of new diagnoses of sarcoidosis in England stratified by age diagnosis and year of diagnosis.

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Figure 3. 18F-FDG PET-CT showing diffuse uptake affecting the liver, spleen, bone marrow, left kidney and lymph nodes above and below the diaphragm. Image used with consent of the patient.

Pro on

Table 1: Examples of markers of end-organ damage, stratified by organ system.

Organ system	Markers of end-organ damage
Cardiovascular	Heart failure Arrhythmia Heart block Implantable cardiac device insertion Pacemaker insertion Cardiac transplantation
Respiratory	Respiratory failure Long-term oxygen therapy Lung transplantation
Neurological	Stroke Transverse myelitis Meningitis Seizure disorder
Ocular	Visual loss Glaucoma surgery (as a complication of chronic uveitis)
Renal	Hyperkalaemia Renal failure Renal transplantation
Liver	Liver transplantation
Endocrine	Hypercalcaemia Hypopituitarism Adrenal crisis
Musculoskeletal	Pathological fracture

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Table 2: Predictors of severe disease and justification for their inclusion in Phase 2 of the Project.

Predictor	Justification for predictor
Age	Our preliminary data shows an increasing incidence of sarcoidosis in older age groups (figure 1).
Gender	Our preliminary data shows an increasing incidence in males (figure 1).
Ethnicity	In American and Scandinavian populations, Black ethnicity is a risk factor for disease severity[7,16,17]. This has not been investigated or quantified in a UK population.
Geographical region, including 13 regions of the UK	This has not been studied in a UK population but has the potential to identify geographical locations of need, with implications on healthcare resource planning.
Socio-economic deprivation, using index of multiple deprivation	In American and Scandinavian populations, social deprivation is a risk factor for disease severity[7]. This has not been investigated or quantified in a UK population.
Organ Involvement	Multiorgan sarcoidosis is linked to a poor outcome. In a Japanese cohort, involvement of three or more organs at presentation suggested a substantially increased risk for poor outcome[32].
Comorbidities including obesity, hypertension, diabetes, asthma and COPD	The co-existence of co-morbidities in sarcoidosis patients has been shown to correlate with poor outcomes in American studies[8].

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Table 3: Clinical outcome data to be collected during Phase 3 of the Project. The finalised list of clinical data will be agreed upon following discussions with stakeholders and patient representatives.

Clinical Data	Measurement tool	Description
Symptom burden	Kings Sarcoidosis Questionnaire[33]	This questionnaire is split into five sections; general health, lungs, medication, skin and eyes
	MRC breathlessness scale[34]	This scale ranges from 0 to 4 and is used to assess the degree of functional disability due to shortness of breath.
	Extrapulmonary organ severity tool[35]	This tool assesses individual organ involvement in patients with sarcoidosis, excluding lung involvement.
Serum biomarkers	Angiotensin converting enzyme (ACE)	ACE is a serum biomarker that is widely used in the diagnosis and monitoring of sarcoidosis. Its level correlates with the amount of whole-body granuloma formation. ACE has a sensitivity of 22%–86% and a specificity of 54%–95% for detecting active sarcoidosis[36].
	Soluble interleukin-2 receptor (sIL-2R)	sIL-2R is a serum biomarker that is used in the diagnosis and monitoring of sarcoidosis. A meta-analysis has estimated that the sensitivity and specificity of sIL-2R levels for active sarcoidosis are 0.88 and 0.87 respectively[36].
Clinician decision for immunomodulatory therapies	Prescription for corticosteroids +/- immunomodulatory therapies	Data will be collected on corticosteroid prescription and doses. We will also collect data on the prescription of second- and third-line immunosuppressive agents
End-organ damage	Summarised in Table 1	
Hospitalisation	Hospitalisation rates	All-cause hospitalisation rates will be measured.
Mortality	All-cause mortality	All-cause mortality rates will be measured.
Health-related quality of life	Sarcoidosis Health Questionnaire[37]	This is a health-related quality-of-life instrument that contains questions relating to three areas; Daily Functioning, Physical Functioning, and Emotional Functioning

Pre-submission

STANFORD UNIVERSITY
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June 26, 2024

Re: Letter of Support for Fellowship Application for the "PAROS Study: Predicting Risk for Adverse Outcome in Sarcoidosis" by Dr. Kathryn Biddle

Dear Fellowship Committee,

I am delighted to provide a letter of support for Dr. Kathryn Biddle's fellowship application. I welcome her plans to perform an in-depth analysis of risk factors for adverse outcomes in sarcoidosis and feel that her research proposal is articulate, well-designed, and impactful.

I would be very happy to facilitate a multi-week placement at our sarcoidosis clinic at Stanford University. This is a world-leading multi-disciplinary unit comprised of experts from a wide range of specialities. Our center leads multiple clinical trials in sarcoidosis, providing an opportunity for Dr. Biddle to gain exposure to trial conduct in an overseas setting. Going forward, this will provide an opportunity to build upon international collaboration between our academic sites.

In conclusion, I fully endorse Dr. Biddle's research proposal and look forward to welcoming her to Stanford University.

Sincerely,

Matthew C. Baker M.D., M.S.
Clinical Chief and Assistant Professor
Division of Immunology and Rheumatology
Stanford University

Pre-submission

King's College Hospital
NHS Foundation Trust

Dear Fellowship Committee,

Letter of Support for Fellowship Application for the "PAROS Study: Predicting Risk for Adverse Outcome in Sarcoidosis" by Dr Kathryn Biddle

I welcome the plans by Dr Kathryn Biddle to undertake a comprehensive analysis of the risk factors for adverse outcomes in patients with sarcoidosis. I have discussed the project with the applicant and her supervisors and am very happy to collaborate.

The management of sarcoidosis can be challenging. There is a poor understanding of risk factors for end-organ damage and few randomised controlled studies exist. Through my involvement in the European Respiratory Consensus guidelines for the treatment of pulmonary sarcoidosis, I acknowledge the lack of evidence underpinning current guidelines.

Dr Biddle's research proposal has the potential to change the way we risk-stratify patients with sarcoidosis. By providing objective measures of disease severity, her research has the potential to optimise clinical trial design and improve trial efficiency. From a clinical perspective, the identification of patients with high-risk disease will improve patient management and guide intelligent resource allocation.

Kings College Hospital hosts one of the UK's largest sarcoidosis centres and will provide world-class clinical and academic support for all aspects of Dr Biddle's proposal.

As a respiratory physician with a research and clinical interest in sarcoidosis, I would be delighted to offer my expertise during the course of this exciting project and look forward to collaborating.

Yours sincerely,

Professor Surinder Birring

Professor of Respiratory Medicine / Consultant Respiratory Physician

Lead, King's Sarcoidosis Service

King's College Hospital

London

Pre-submission

King's College Hospital NHS
NHS Foundation Trust

1st July 2024

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Dear Fellowship Committee,

Letter of Support for Fellowship Application for the "PAROS Study: Predicting Risk for Adverse Outcome in Sarcoidosis" by Dr Kathryn Biddle

I am writing to confirm my commitment to collaborate with Dr Biddle on her Fellowship Application.

PET-CT is a useful imaging modality in the diagnosis, work-up and monitoring of sarcoidosis. It can characterise disease burden, detect occult organ involvement and monitor response to treatment. Despite this, there are no objective grading scores for sarcoidosis activity on PET-CT and the potential applications of PET-CT in sarcoidosis have not been fully explored.

I fully support the plans by Dr Biddle to analyse the imaging risk factors for disease severity and progression in sarcoidosis. I have discussed this project with Dr Biddle and I am happy to offer hands-on radiology expertise, to support her in reading and labelling the PET-CT scans.

I wish her the best of luck with the application.

Yours sincerely,

Dr Marko Berovic
Consultant Radionuclide Radiologist

King's College Hospital
NHS Foundation Trust

Dear Fellowship Committee,

Letter of Support for Fellowship Application for the "PAROS Study: Predicting Risk for Adverse Outcome in Sarcoidosis" submitted by Dr Kathryn Biddle

I am delighted to write a letter of support for Dr Kathryn Biddle to perform a detailed analysis of the risk factors for adverse outcomes in sarcoidosis. Her research proposal is well-designed and has the potential to generate high impact outputs.

Cardiac sarcoidosis carries a high morbidity and mortality rate. The advent of newer imaging modalities, including PET-CT, have shown a higher prevalence of cardiac involvement in sarcoidosis than previously thought. Despite its clinical importance, the optimal management of cardiac sarcoidosis is not well understood. The management of cardiac sarcoidosis in the context of multi-system disease is poorly defined and requires a multi-disciplinary approach.

Dr Biddle's research outputs have the potential to change the way we prognosticate and manage patients with multi-system sarcoidosis. I agree that a multi-disciplinary research approach is important and I welcome Dr Biddle's multi-speciality collaborative proposal.

To this end, I would be delighted to offer my expertise in cardiac imaging and cardiac sarcoidosis to this project. I fully support this research proposal and look forward to collaborating.

Yours sincerely,

Dr Dan Sado, BM, BSc, MRCP, MD

Consultant in Cardiology, Kings College Hospital

Honorary Senior Lecturer, Kings College London

Pre-submission

Appendix #3: Gantt Chart.pdf

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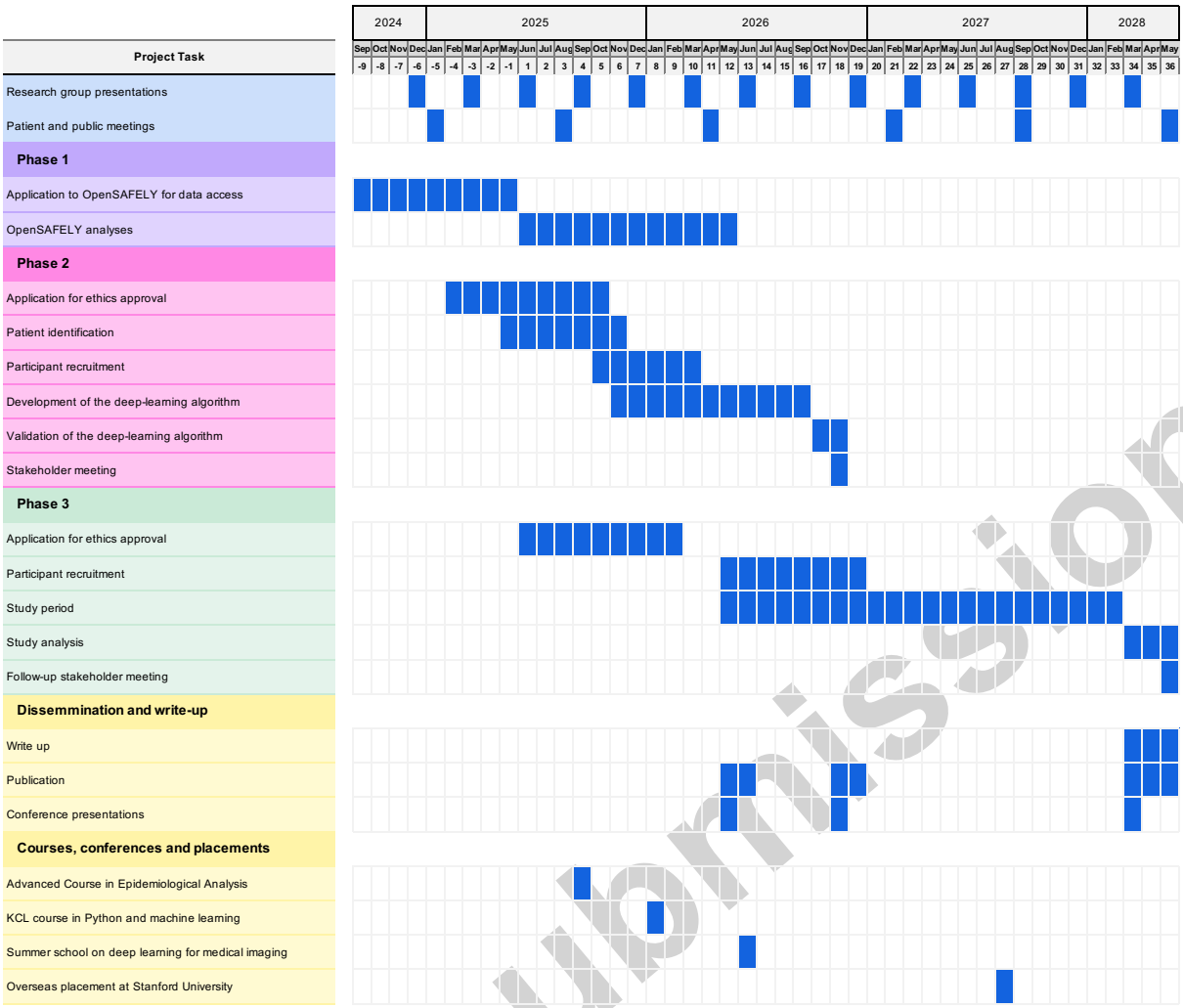
Gantt Chart



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Appendix #4: Funder Export_346985_SoECAT_PAROS Study.xlsx

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